

Assignment 2: Design a Logical Model and Advanced SQL

Assignment2.md

Section 1:

Prompt 3

The architecture for the CUSTOMER_ADDRESS table that retain changes is type 2 and the other that will overwrite the changes are type 1.

In database design, especially in the context of data warehousing and dimensional modeling, type 1 and type 2 represent different strategies for handling slowly changing dimensions (SCDs).

Overview

Type 1:

Direct updates on existing record and replace old data with new value.

Best use case for data element where historical accuracy is not important.

Type 2:

Inserts new row for each change while retaining old record. Addition fields such as start/end date, flag, or version number are needed to maintain historical views of data.

Best use case for where historical views are important to maintain and analyze.

The table below compares their differences

Feature	Type 1 – Overwrite Existing Row	Type 2 – Add New Row with History
What Happens	Updates existing record in place	Inserts new record, keeps old one
History Preserved	No	Yes
Extra Columns	None needed	Start/End Date, Flag, Version Number

Use Case	When history isn't important	When historical accuracy and analysis are needed
Storage Impact	Low (single row per entity)	Higher (multiple rows per entity)